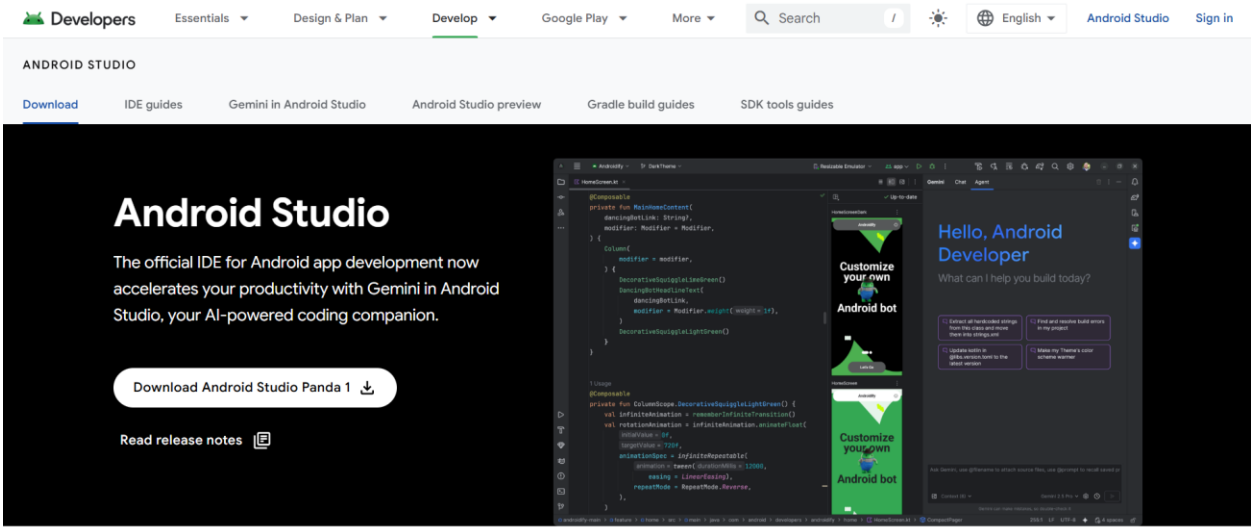
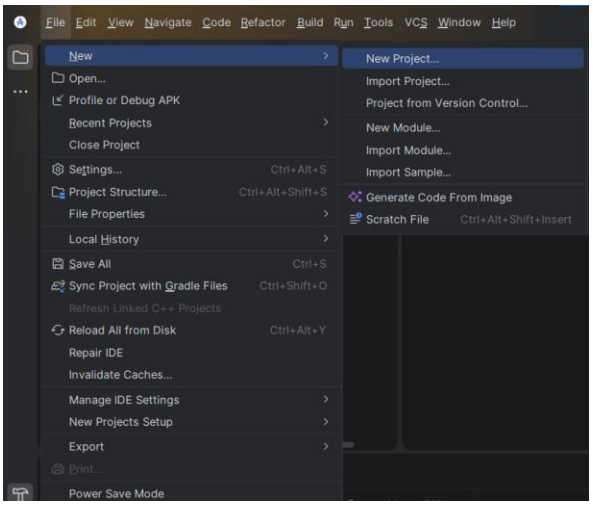
Lab 1: Android Studio, Activity, Intent and GUI

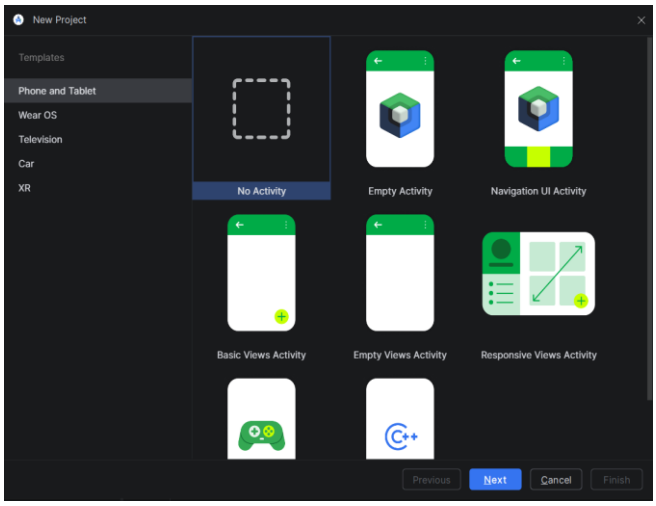
I. Android Studio

Download Android Studio: https://developer.android.com/studio

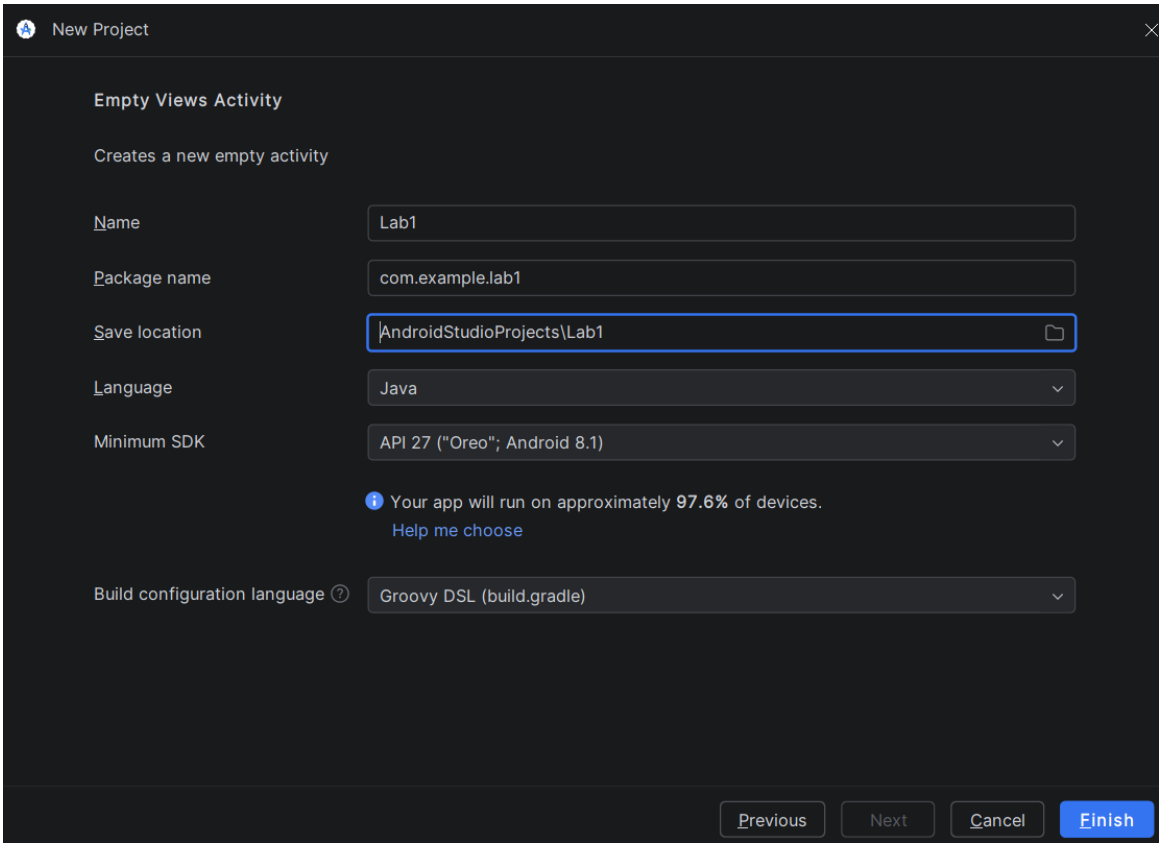


After installing successfully, you can create a new project (choose “Empty Views Activity”):



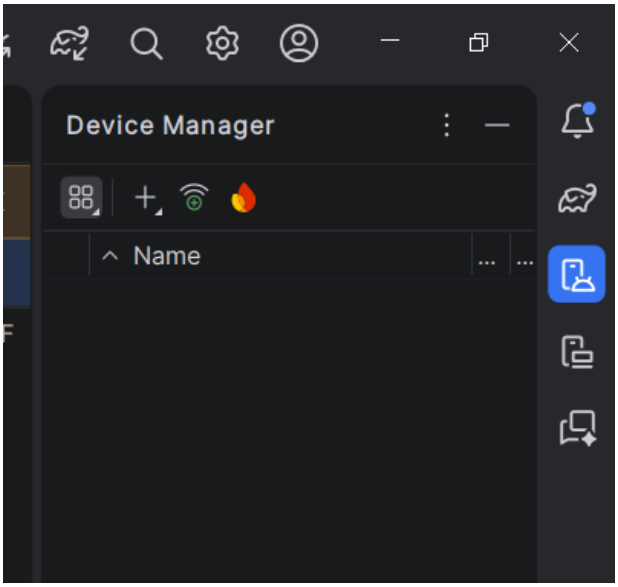


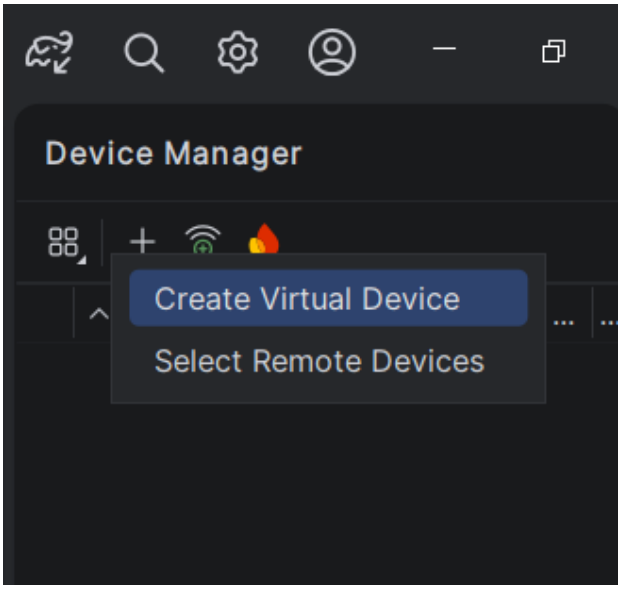
Set the project name:

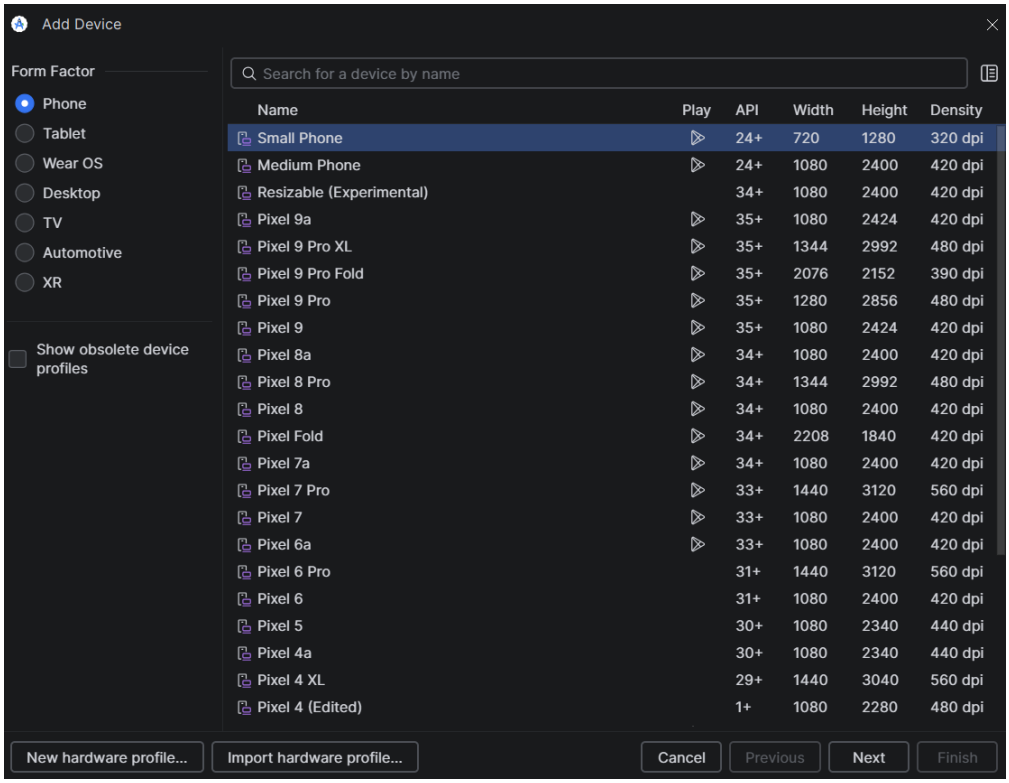


Create a Virtual Device:

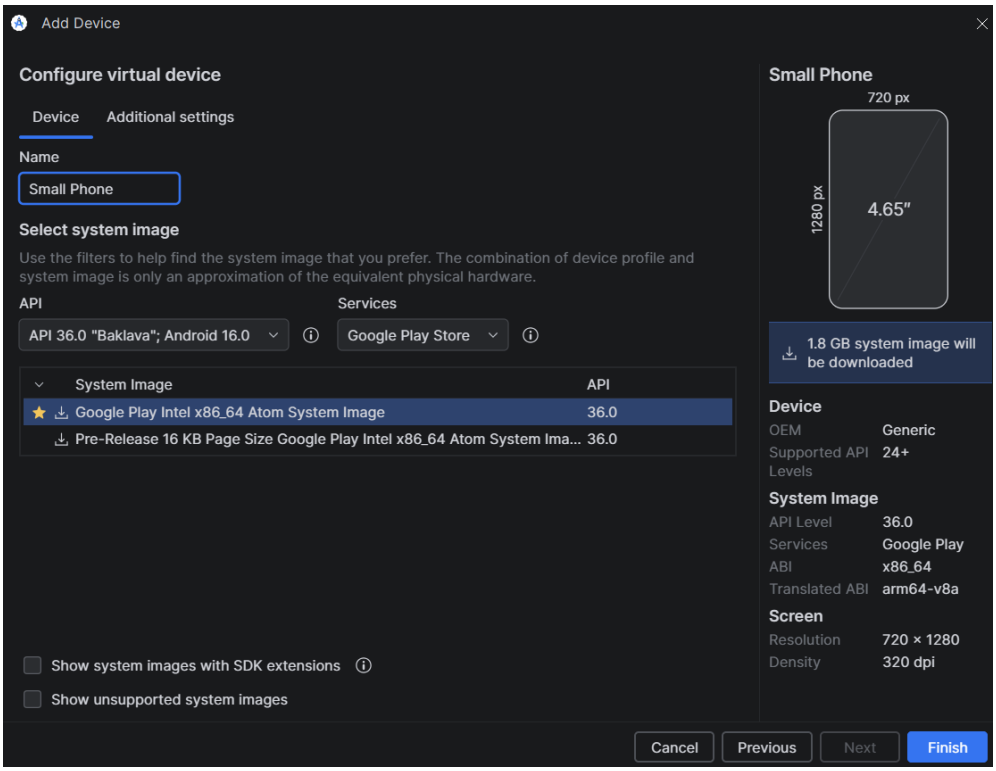
Go to **Device Manager**, select **Create Virtual Device**, then choose the device type (here we choose **Phone**)

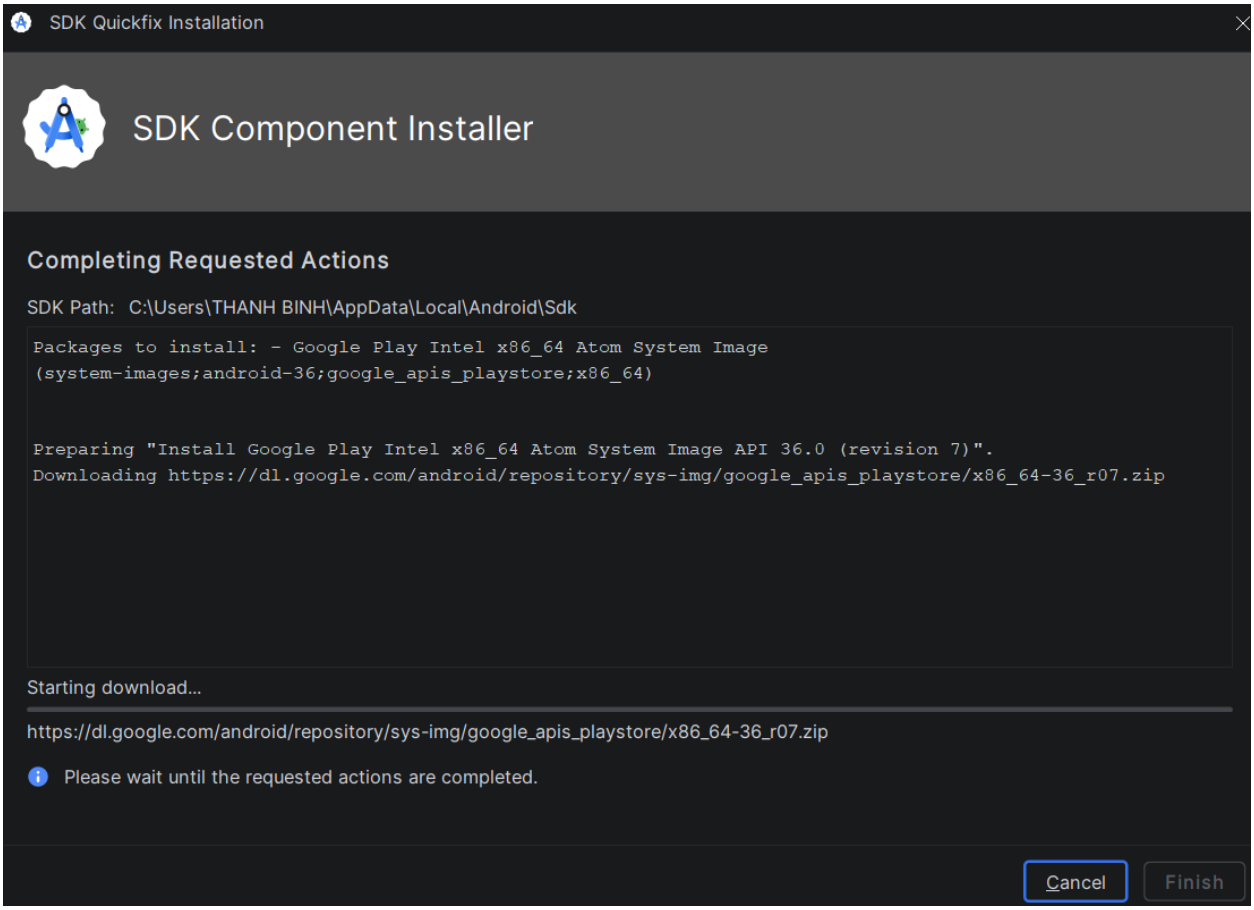






Check the device information and proceed to create the virtual device:





II. Activity

An Activity is a component in Android that represents a single screen with a user interface.

Each Activity:

- Displays a UI to the user.
- Handles user interactions.
- Has its own lifecycle.

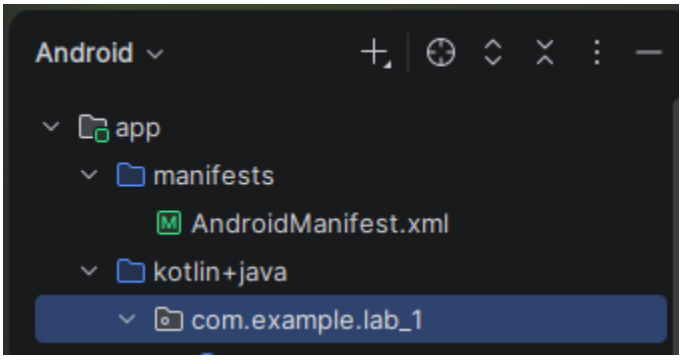
Common lifecycle methods:

- onCreate()
- onStart()
- onResume()
- onPause()

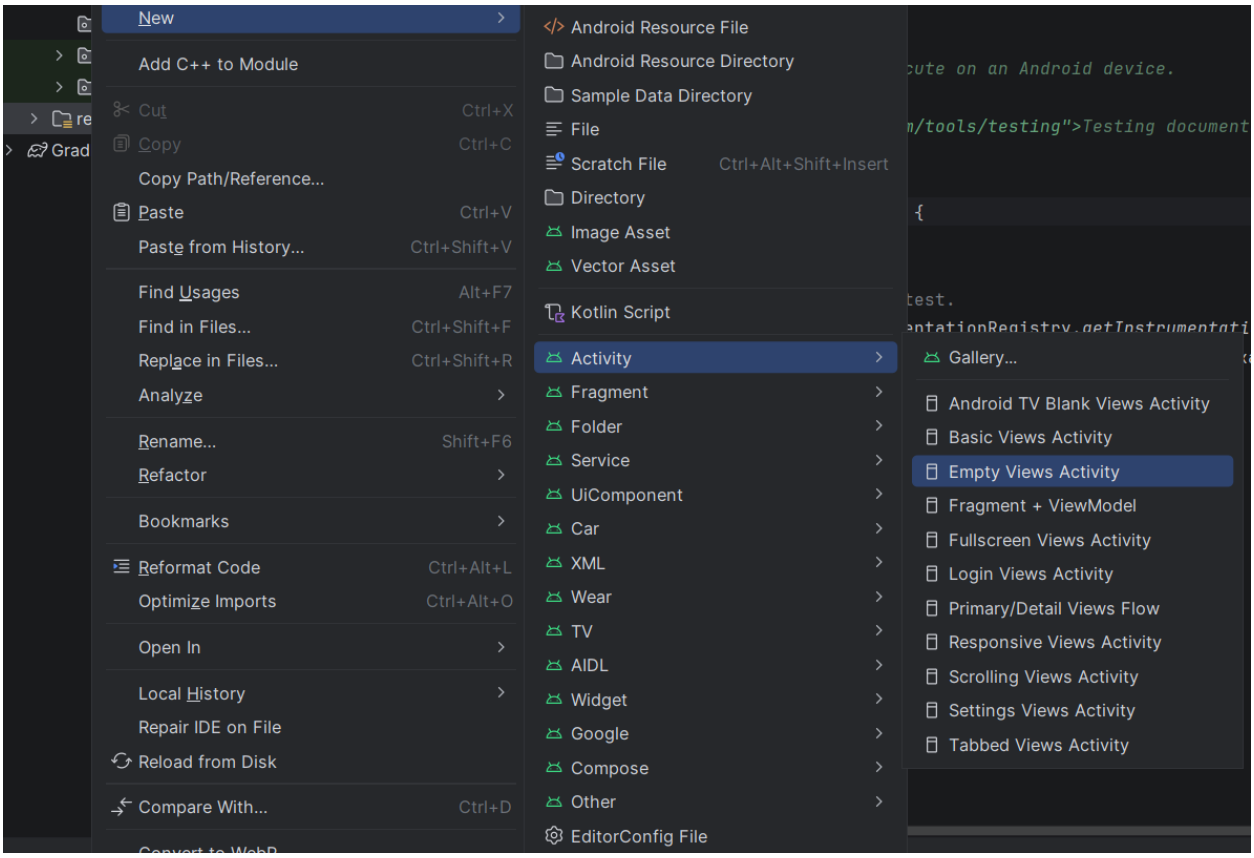
- onStop()
- onDestroy()

Example:

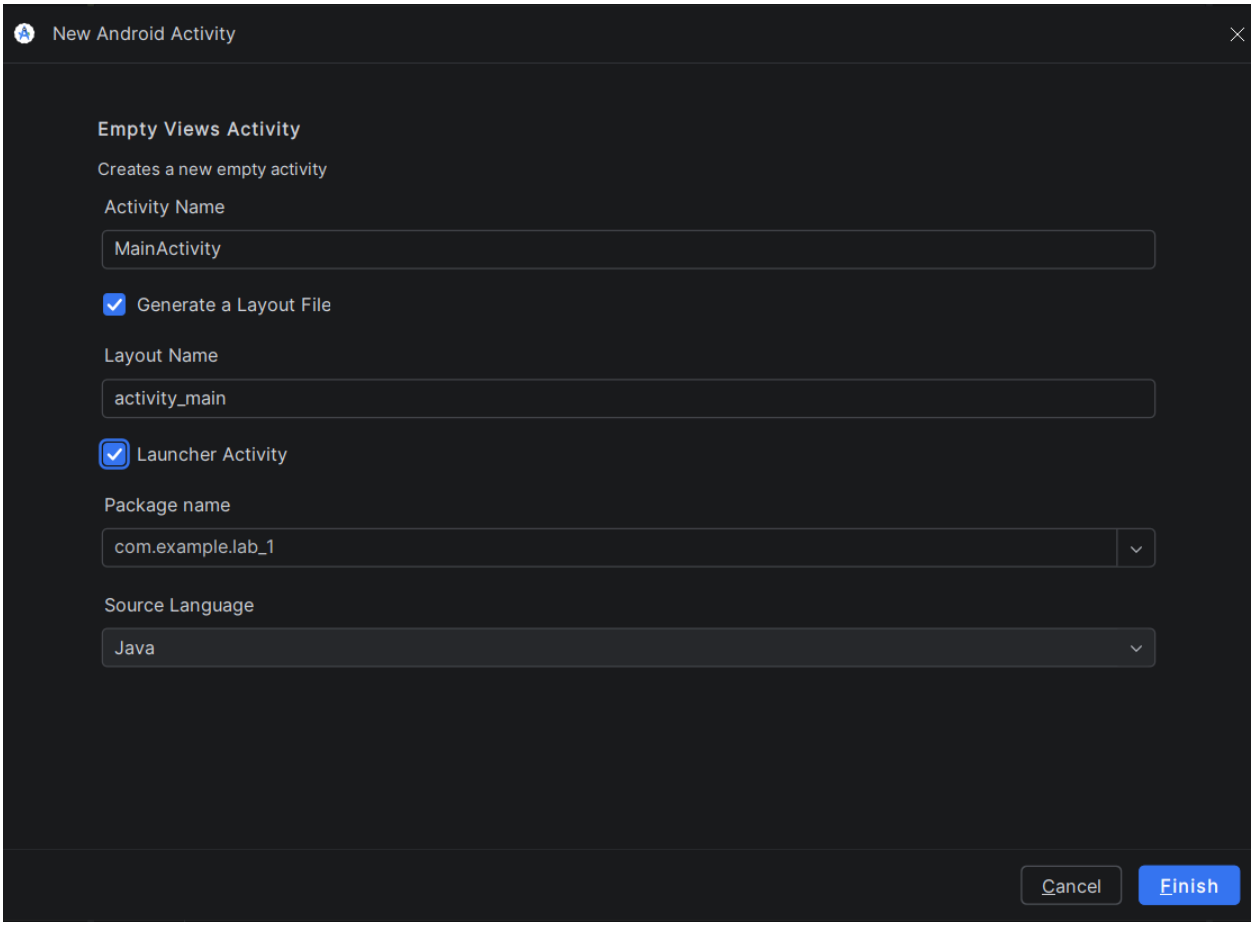
After creating a new project (**Empty Views Activity**), go to the path: *app.com.example.lab_1*



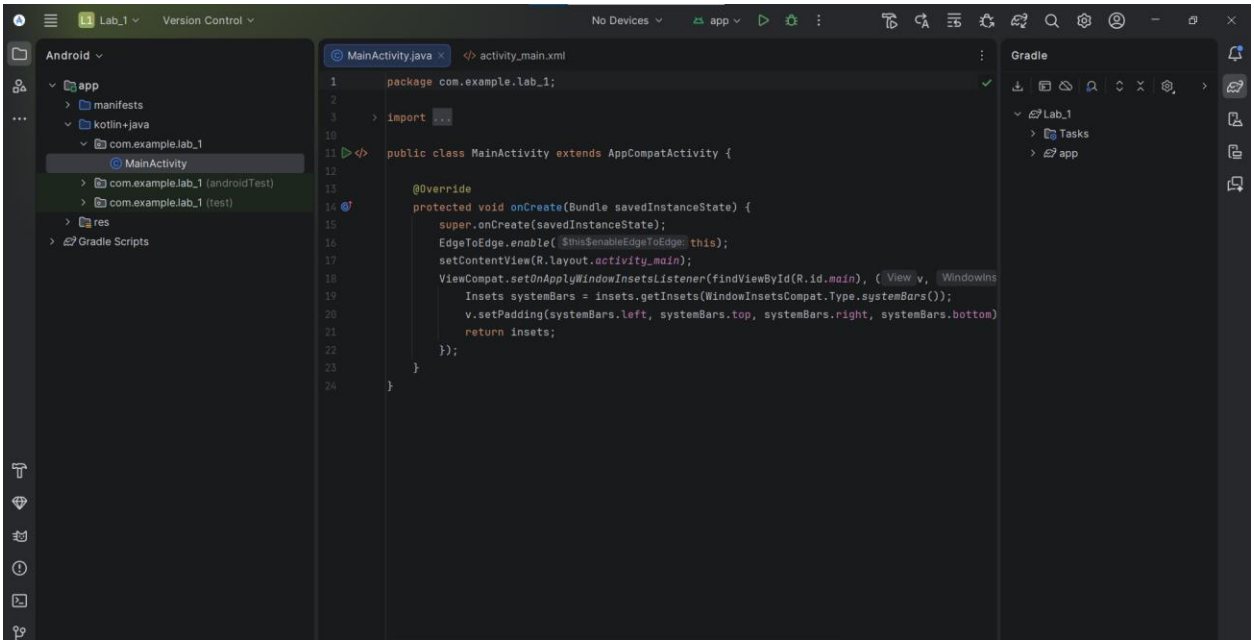
Create a new **Empty Views Activity**.



Set this Activity as **MainActivity**:



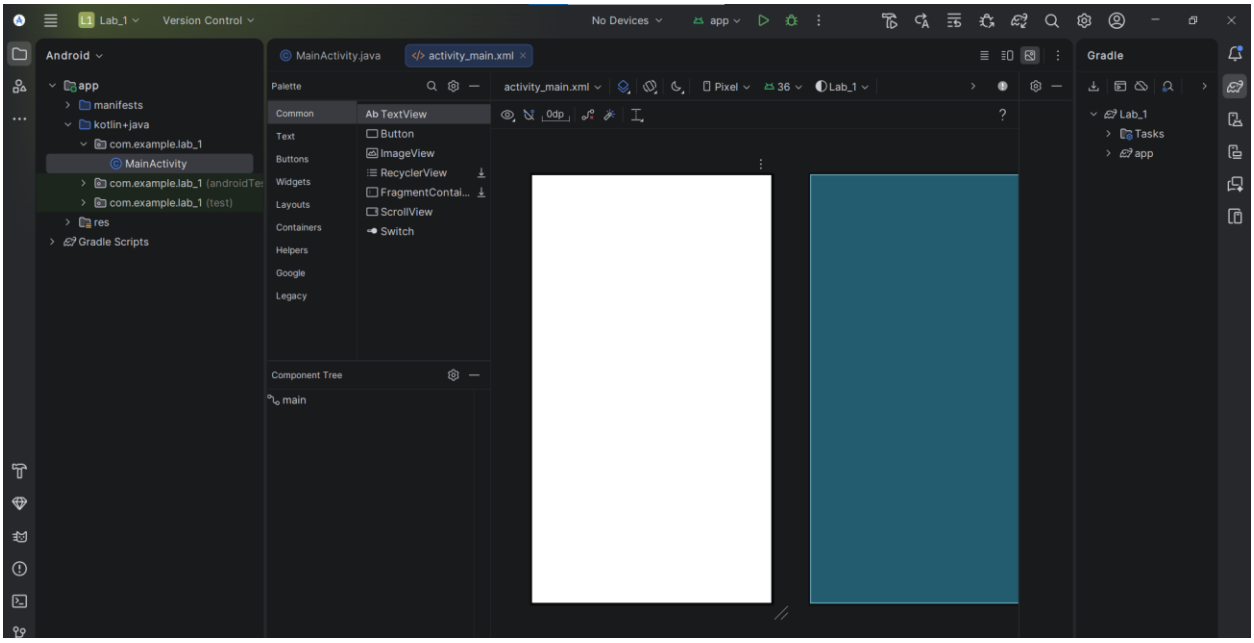
Default code after initializing the Activity:



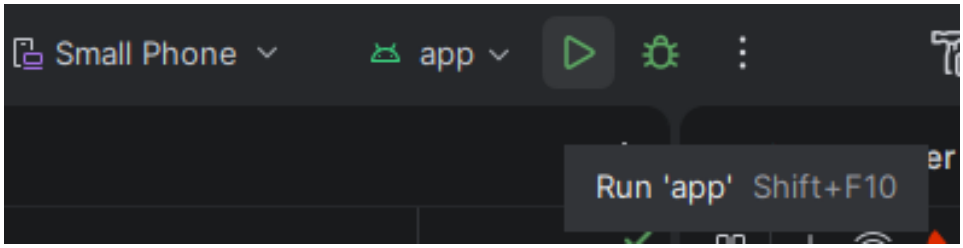
File *activity_main.xml* - this file is used to design the UI layout for the Activity:

```xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
</ScrollView>
```

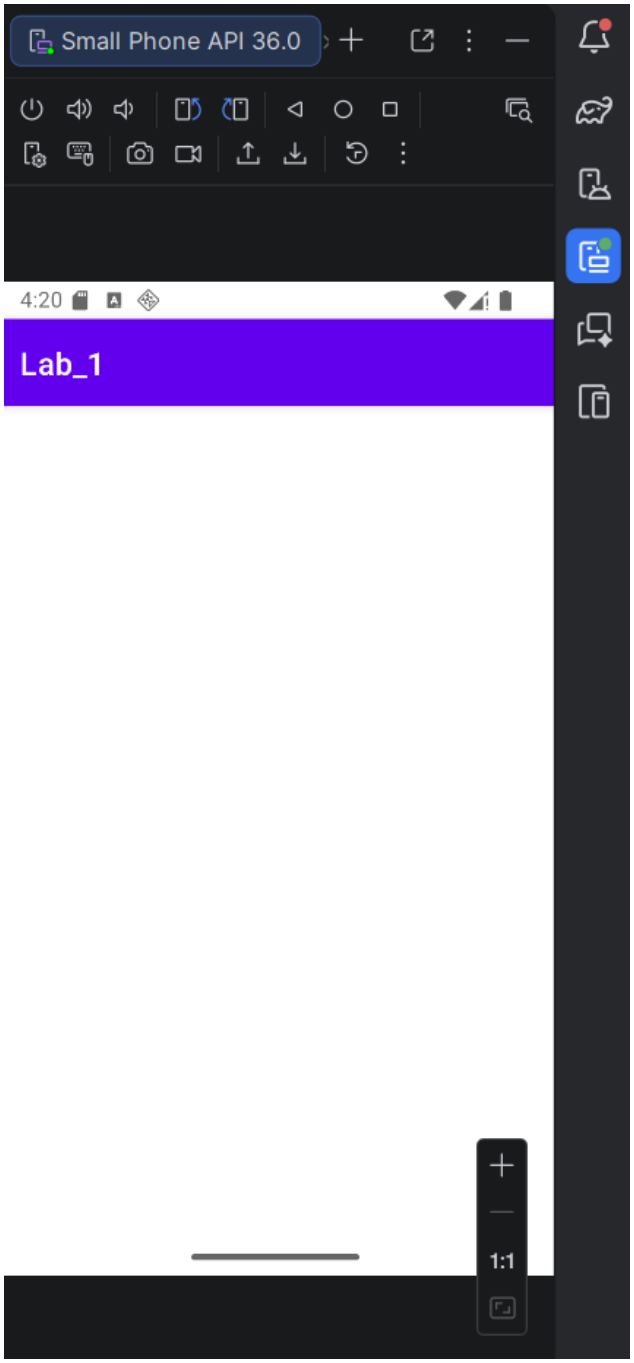
Interface corresponding to the XML file:



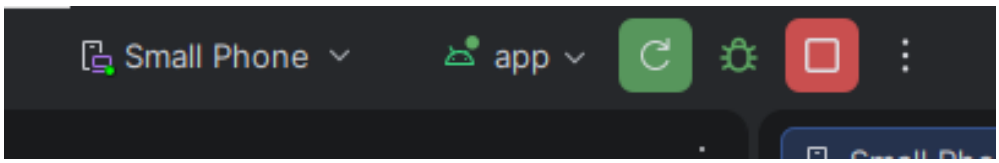
To run the application, select **“Run ‘app’”**:



The application after running on the virtual device:



After making changes to the code, you can run the application again to see the updates:



II. Intent

An **Intent** is a messaging object used to **communicate between Android components**.

Intents are used to:

- Start another Activity
- Pass data between Activities
- Open another application

There are two types of Intents:

Explicit

An Explicit Intent specifies the exact Activity to start.

Used mainly for navigation within the same application.

Implicit

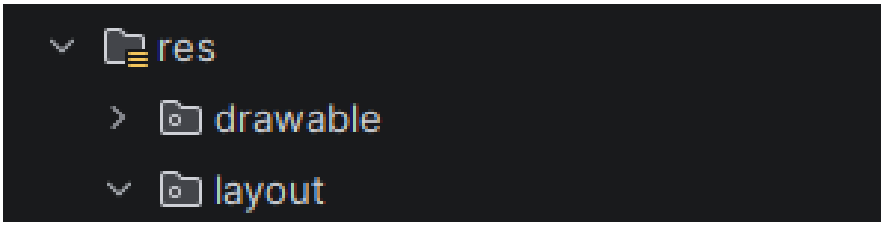
An Implicit Intent does not specify a target Activity.
Android automatically finds an application capable of handling the request.

Example code:

File MainActivity:

```java
public class MainActivity extends AppCompatActivity {

    2 usages
    EditText edtName;
    2 usages
    Button btnSend;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btnWeb = findViewById(R.id.btnOpenWeb);

        btnWeb.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

                Intent intent = new Intent(Intent.ACTION_VIEW);
                intent.setData(Uri.parse( uriString: "https://www.google.com"));
                startActivity(intent);

            }
        });

        edtName = findViewById(R.id.edtName);
        btnSend = findViewById(R.id.btnSend);

        btnSend.setOnClickListener( View v -> {

            String name = edtName.getText().toString();

            Intent intent = new Intent( packageContext: MainActivity.this, SecondActivity.class);
            intent.putExtra( name: "username", name);

            startActivity(intent);
        });
    }
}
```

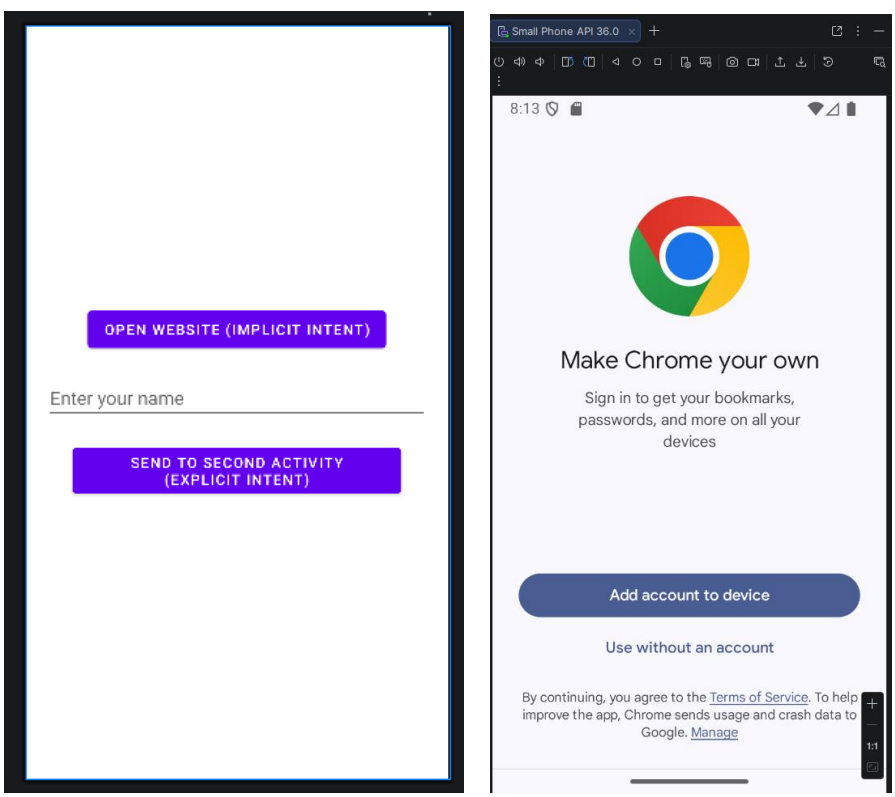
activity_main.xml and activity_second.xml are located in res/layout:



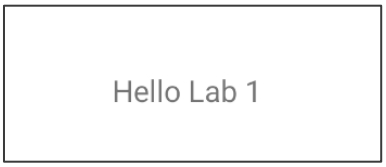
```xml
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btnOpenWeb"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp"
        android:text="Open Website (Implicit Intent)" />

    <EditText
        android:id="@+id/edtName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter your name"/>

    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send to Second Activity
        (Explicit Intent)"
        android:layout_marginTop="20dp"/>

</LinearLayout >
```

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout  xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".SecondActivity"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/txtResult"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="24sp"/>


</LinearLayout >
```

When running the application, if you click **Button 1**, the result will open a **web browser** with the specified URL:



If you enter information and press Button 2, the entered data will be passed to SecondActivity:



III. GUI and Widget

A **Graphical User Interface (GUI)** is the visual interface that allows users to interact with an application through graphical elements instead of text commands.

Examples of GUI elements:

- Buttons
- Text fields
- Images
- Menus

In Android:

- GUI layouts are usually defined using **XML**.
- Application logic is written in **Java or Kotlin**.

Widget

Widgets are pre-built user interface components provided by the Android framework.

They are used to build application interfaces quickly and efficiently.

Common Android Widgets:

Widget	Function
TextView	Displays text
EditText	Accepts user input
Button	Clickable button
ImageView	Displays images
CheckBox	Multiple-choice option
RadioButton	Single-choice option
Spinner	Dropdown menu
ListView	Displays a scrollable list

Example code:

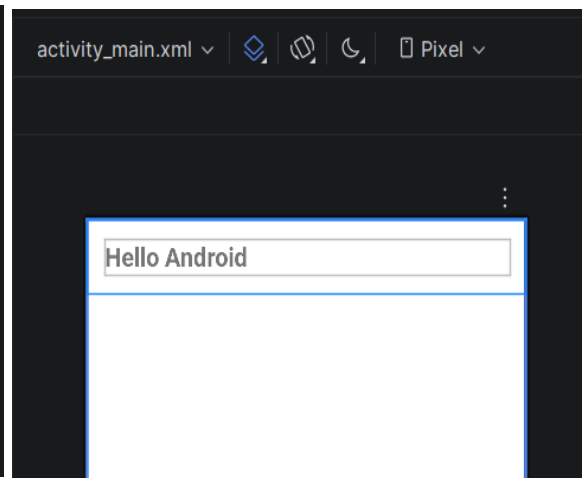
File main_activity.xml:

```xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
</ScrollView>
```

Then, add widgets step by step in <ScrollView>:

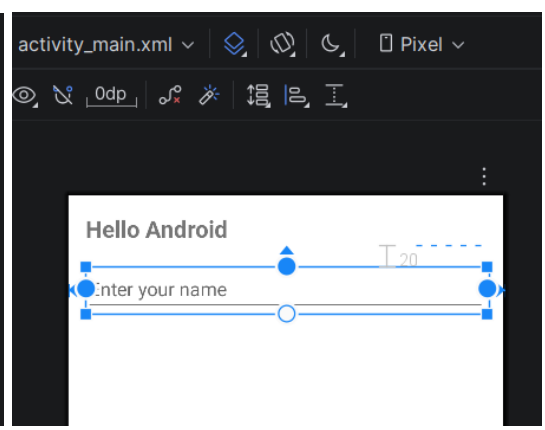
1. TextView:

```xml
<TextView
    android:id="@+id/txtHello"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Hello Android"
    android:textSize="22sp"
    android:textStyle="bold"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    tools:ignore="HardcodedText" />
```



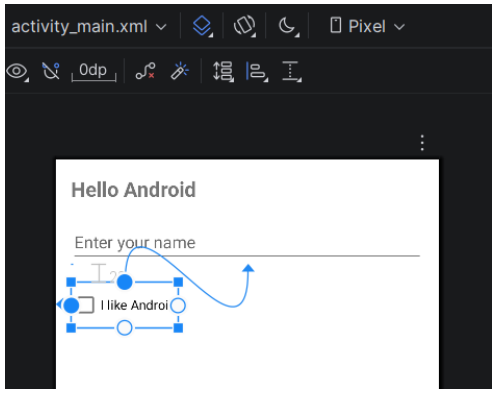
2. EditText:

```xml
<EditText
    android:id="@+id/edtName"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:hint="Enter your name"
    android:layout_marginTop="20dp"
    app:layout_constraintTop_toBottomOf="@id/txtHello"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraint End_toEndOf="parent"
    tools:ignore="Autofill,HardcodedText,TextFields" />
```



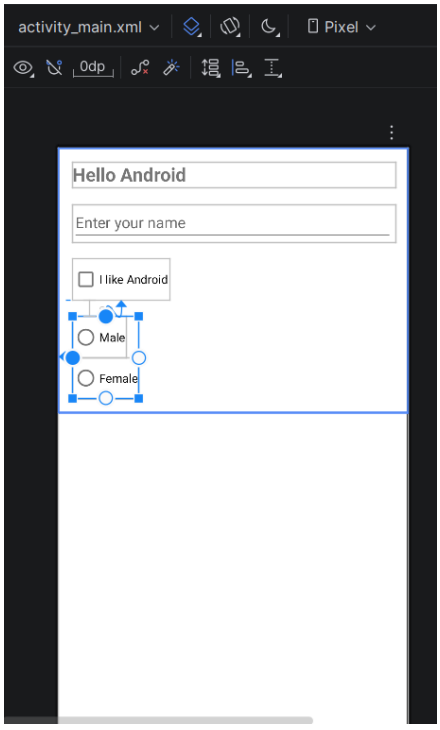
3. CheckBox:

```
<CheckBox
    android:id="@+id/chkAndroid"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="I like Android"
    android:layout_marginTop="20dp"
    app:layout_constraintTop_toBottomOf="@id/edtName"
    app:layout_constraintStart_toStartOf="parent"
    tools:ignore="HardcodedText" />
```



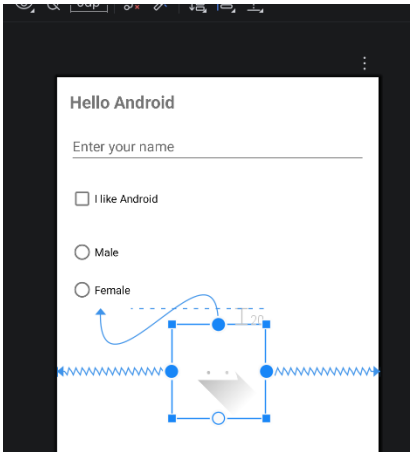
4. RadioGroup:

```
<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="20dp"
    app:layout_constraintTop_toBottomOf="@id/chkAndroid"
    app:layout_constraintStart_toStartOf="parent">

    <RadioButton
        android:id="@+id/radioMale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male"
        tools:ignore="HardcodedText" />

    <RadioButton
        android:id="@+id/radioFemale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female"
        tools:ignore="HardcodedText" />
</RadioGroup>
```



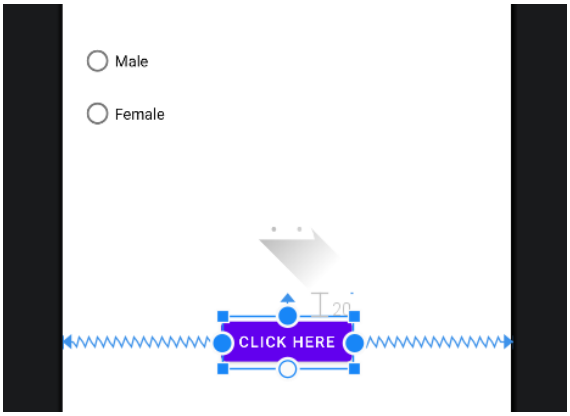
5. ImageView:

```
<ImageView
    android:id="@+id/imgLogo"
    android:layout_width="120dp"
    android:layout_height="120dp"
    android:src="@drawable/ic_launcher_foreground"
    android:layout_marginTop="20dp"
    app:layout_constraintTop_toBottomOf="@id/radioGroup"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    tools:ignore="ContentDescription" />
```



6. Button:

```
<Button
    android:id="@+id/btnSubmit"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Click Here"
    android:layout_marginTop="20dp"
    app:layout_constraintTop_toBottomOf="@id/imgLogo"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraint End_toEndOf="parent"
    tools:ignore="HardcodedText" />
```



IV. Exercise:

Exercise 1: To-Do List application

Functional Requirements

1. Users can add a new task using an EditText and Button.
2. Tasks are displayed in a ListView / RecyclerView.
3. Each task has:
    - Task name
    - Checkbox to mark Completed
4. When a task is completed:

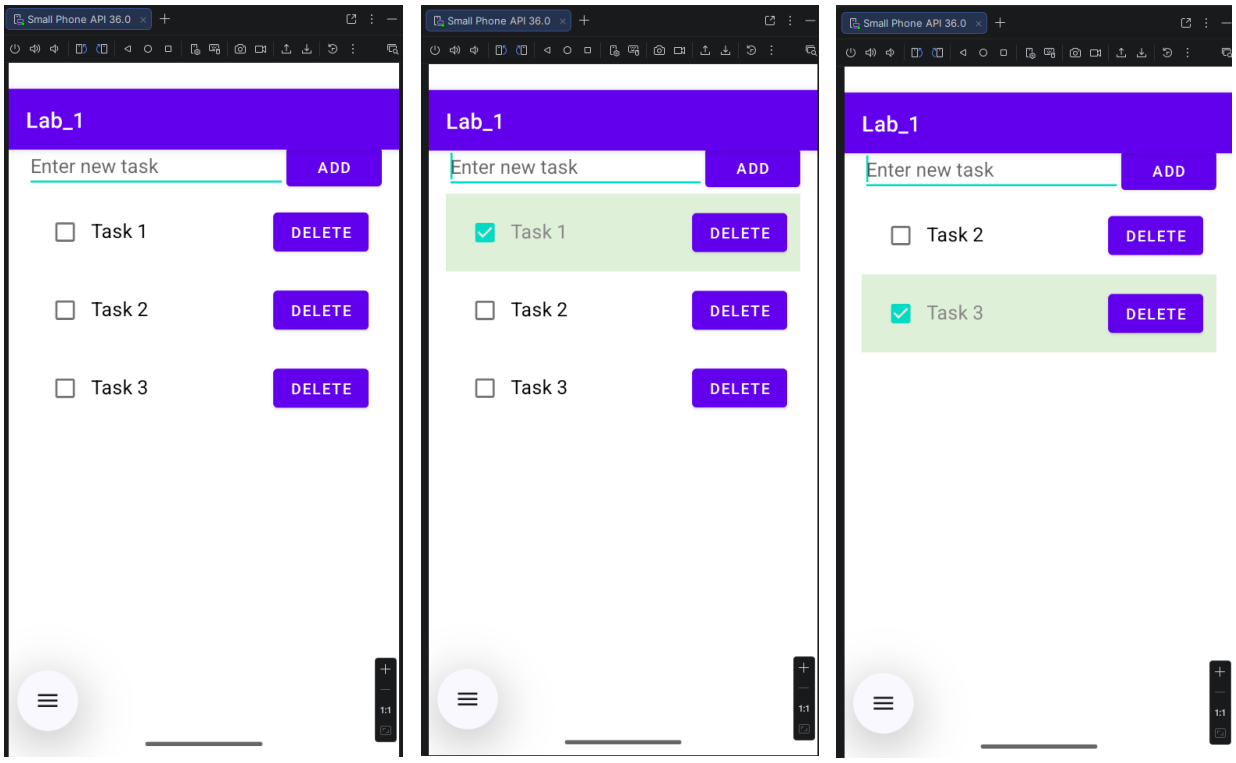
    - The text color changes to gray
    - The background changes to light green
5. Users can delete a task.

UI Requirements

The interface must contain:

- EditText for entering task
- Button to add task
- RecyclerView to display tasks
- Each item includes:
    - Checkbox
    - Task title
    - Delete button

Illustration result



Exercise 2: Student Grade Management App

Objective: Build an application to manage student grades.

Screens

MainActivity - Display the list of students

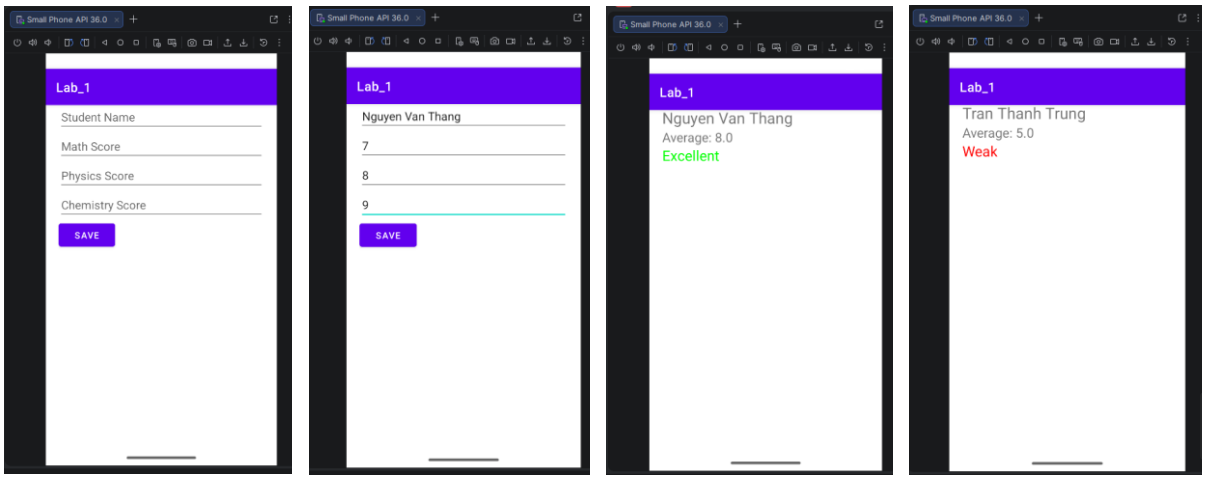
AddStudentActivity - Add a new student

DetailActivity - View detailed scores

Features

- Add a student
- Input scores for 3 subjects
- Calculate the Average Score
- Automatically classify student performance
- Change text color based on performance level

Illustration result



Exercise 3: Shopping Cart App

Objective: Simple shopping cart

Screens

ProductListActivity - Show products

CartActivity - Cart items

CheckoutActivity - Payment

Features

- Add to cart
- Remove product
- Quantity update
- Calculate total price

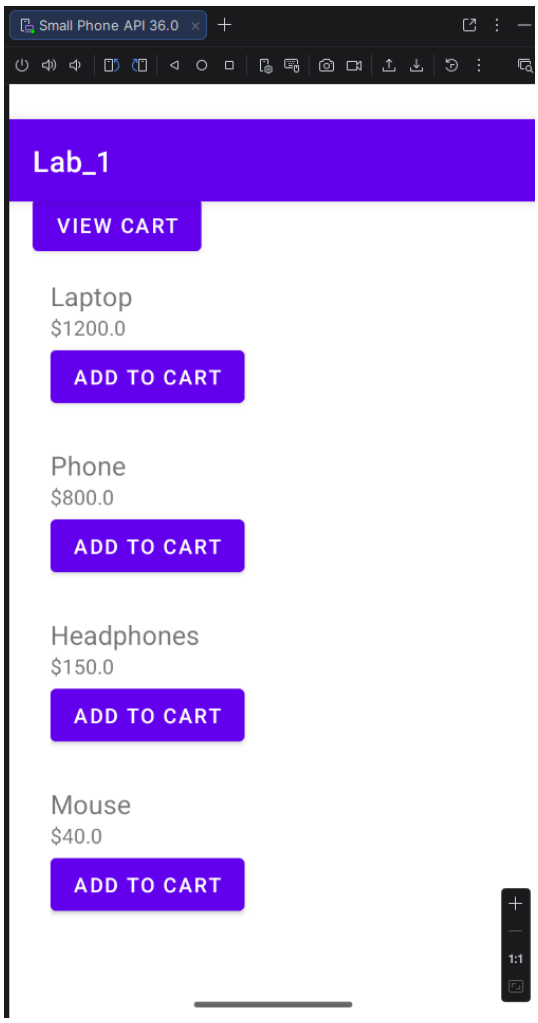
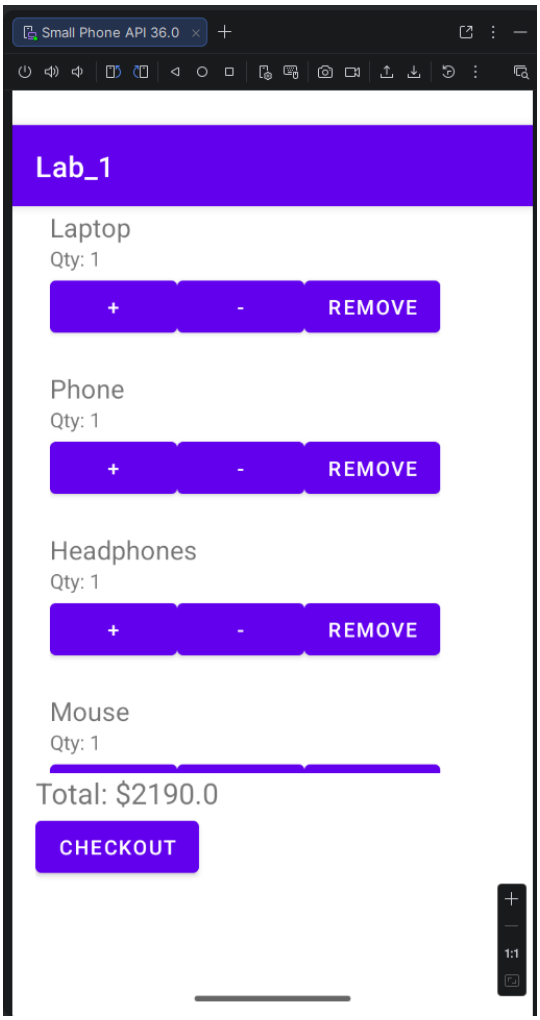



Illustration result